Introspection Dynamics with Mutation in Additive Games

Harry Foster[1], Vince Knight[1,*], and Sebastian Krapohl[2]

[1]School of Mathematics, Cardiff University
[2]Faculty of Social and Behavioural Sciences, University of Amsterdam
[*]Corresponding author: knightva@cardiff.ac.uk

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Abstract

Cooperation in heterogeneous groups, where individuals differ in resources, productivity, and behavioural responsiveness, underpins collective action across social and biological systems. Introspection dynamics is a learning rule suited to such asymmetric settings: at each time step a randomly selected player compares their current payoff to the payoff an alternative action would have given, and switches with a probability increasing in the difference. Couto and Pal showed that for *additive* games, those in which the payoff difference a player evaluates when considering a switch is independent of the other players' actions, the stationary distribution of introspection dynamics is a product measure. We extend this result to introspection dynamics with *mutation*, where a selected player switches to cooperation or to defection with payoff-independent, player-specific probabilities, and we allow player-specific selection intensities. We show that the product structure is preserved, and we obtain an explicit formula for each player's long-run cooperation probability in terms of their own payoffs, selection intensity, and mutation probabilities alone. We consider the heterogeneous public goods game, where N players may differ in their contributions, public goods multipliers r_i, and selection intensities β_i; the long-run cooperation probability admits a closed form with N terms rather than 2^N. Several structural consequences follow: a player-specific cooperation threshold at $r_i = N$ under symmetric mutation, a neutral-drift regime in which cooperation is governed entirely by mutation bias, and a mutation-selection balance in which aggregate cooperation is affine in the mutation rate, interpolating between selection and neutrality. Mutation also regularises the strong-selection limit $\beta_i \to \infty$, where the mutation-free dynamics degenerate.

1 Introduction

The public goods game is a canonical model of cooperation in evolutionary game theory: players decide whether to contribute to a common resource, which is multiplied and redistributed, creating a tension between individual incentive and collective benefit [7]. Classical analyses assume a homogeneous population, but real groups are heterogeneous: players differ in their productivity, their contributions, and how sensitively they respond to payoff differences [6].

A natural learning rule for such settings is *introspection dynamics*, introduced in [1]: at each time step a player compares their current payoff to the payoff they *would have received* had they played an alternative action (keeping all other players fixed), and switches with a probability governed by this difference. Because the comparison is self-referential rather than imitative, introspection applies to arbitrary asymmetric games where role-model imitation is unavailable or unnatural.

In [2] the authors extended introspection dynamics to multiplayer asymmetric games. For a general game the transition rate for player i switching depends on the actions of all other players through the payoff function, so the stationary distribution must be computed by solving a linear system of dimension 2^N (more generally L^N when each player has L actions). They identified an important exception: for *additive* games, in which the payoff difference a player evaluates when switching action is independent of the other players' actions (a property also known as *equal gains from switching*), the stationary distribution factorises as a product measure over the players ([2]; Proposition 2), and for the linear public goods game it admits a closed form ([2]; Proposition 3).

The introspection model of [1, 2] contains no mutation: a selected player switches action solely according to the Fermi rule. Mutation, a random action chosen independently of payoffs, is a standard ingredient of evolutionary and learning dynamics [5] and accounts for exploration, error, and behavioural noise. In this paper we incorporate per-player, per-action mutation into introspection dynamics and show that the additive-game results of [2] persist: the stationary distribution remains a product measure, and each player's long-run cooperation probability has the explicit form $p_i = \phi_i(\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i0}$. We also allow player-specific selection intensities β_i rather than a single global value. Specifically, we extend the product-measure result to mutation (Section 3), record the structural consequences that hold for any additive game (Section 4), and specialise them to the heterogeneous public goods game (PGG), for which additivity holds and the closed-form cooperation probability follows (Section 5).

The paper is organised as follows. Section 2 introduces the framework, the additive-game condition, and introspection dynamics with mutation, noting that whilst the donation-form Prisoner's Dilemma is additive, the Stag Hunt and the general Prisoner's Dilemma are not. Section 3 shows that, with mutation, individual cooperation probabilities are given by (4) and the stationary distribution remains a product measure. Section 4 records the structural consequences that hold for any additive game: a neutral-drift regime, a strong-selection limit, a cooperation threshold, and a mutation-selection balance. Section 5 specialises these to the heterogeneous PGG, establishing additivity and deriving the closed-form cooperation probability. Section 6 concludes.

2 Setup and notation

Consider N ordered players, each choosing from two actions $\{C, D\}$ coded as $\{1, 0\}$. The state space is $S = \{0,1\}^N$ with $|S| = 2^N$. Each player i has a payoff function $f_i : S \to \mathbb{R}$ and player-specific selection intensity $\beta_i > 0$. Writing $\mathbf{a}_{i\to\bar{a}_i}$ for the state with player i's action flipped to $\bar{a}_i = 1 - a_i$, the *payoff difference* for player i at state $\mathbf{a}$ is

$$\Delta f_i(\mathbf{a}) := f_i(\mathbf{a}) - f_i(\mathbf{a}_{i\to\bar{a}_i}).$$

Following [2], a game is *additive* for player i if $\Delta f_i(\mathbf{a})$ depends only on a_i, not on $\mathbf{a}_{-i}$, the actions of all individuals other than i; equivalently, the payoff difference a player evaluates when considering a switch is independent of the co-players' actions, a property also known as *equal gains from switching*. When this holds there is a constant $\delta_i \in \mathbb{R}$ such that

$$\Delta f_i(\mathbf{a}) = (1 - 2a_i)\,\delta_i, \tag{1}$$

where $\delta_i = \Delta f_i\big|_{a_i=0}$ is the payoff difference when player i defects. When $a_i = 0$, (1) follows directly from the definition of δ_i; when $a_i = 1$, $\Delta f_i\big|_{a_i=1} = -\Delta f_i\big|_{a_i=0} = -\delta_i$, giving (1).

Note that not all games are *additive*. For example in [1] two of the two-player, two-strategy games considered to illustrate introspection dynamics are the Prisoner's Dilemma M_1 and the Stag

Hunt M_2 of (2), for $0 < c_1, c_2 < b$.

$$M_1 = \begin{pmatrix} b-c_1, b-c_2 & -c_1, b \\ b, -c_2 & 0, 0 \end{pmatrix} \qquad M_2 = \begin{pmatrix} b-c_1, b-c_2 & -c_1, 0 \\ 0, -c_2 & 0, 0 \end{pmatrix} \tag{2}$$

For M_1 we have $\delta_1 = c_1$ and $\delta_2 = c_2$. However for M_2, writing (a_1, a_2) for the state in which player 1 plays a_1 and player 2 plays a_2, $\Delta f_1((1,1)) = b - c_1$ but $\Delta f_1((1,0)) = -c_1$. The Prisoner's Dilemma (as defined by M_1) is *additive* but the Stag Hunt game (as defined by M_2) is not. Further games considered in [1] are the Volunteer's Dilemma and the Volunteer's Timing Dilemma (a generalisation with more than 2 strategies), neither of which are *additive*.

More generally, the general Prisoner's Dilemma (PD) is parameterised by the reward R, temptation T, sucker payoff S, and punishment P with $T > R > P > S$ and the social-efficiency condition $2R > T + S$:

$$M_3 = \begin{pmatrix} R, R & S, T \\ T, S & P, P \end{pmatrix}.$$

Additivity requires, for each player, the *equality*

$$T - R \;=\; P - S \quad (\text{equivalently } P + R = S + T),$$

that is, the gain from defecting against a cooperator must equal the gain from defecting against a defector. The PD axioms are strict inequalities and do not force this equality; a generic PD therefore fails it. For instance, $T = 5$, $R = 3$, $P = 1$, $S = 0$ satisfies $T > R > P > S$ and $2R = 6 > 5 = T + S$, yet $T - R = 2 \neq 1 = P - S$. The donation form M_1 above is the exceptional case where the equality holds by construction (for player i, both differentials equal c_i).

To define introspection dynamics we introduce the following two components:

- **Fermi function.** Each player i has a personal Fermi function $\phi_i(x) = (1 + e^{\beta_i x})^{-1}$. Since $\beta_i > 0$, the exponential $e^{\beta_i x}$ is strictly increasing in x, so ϕ_i is strictly decreasing with $\phi_i(0) = \frac{1}{2}$ and $\phi_i(x) + \phi_i(-x) = 1$.
- **Mutation.** Each player i has mutation probabilities $\mu_{i0}, \mu_{i1} > 0$ with $\mu_{i0} + \mu_{i1} < 1$, where μ_{i0} is the probability of mutating to cooperation and μ_{i1} is the probability of mutating to defection, regardless of the intended choice of the dynamic.

At each step, one player i is selected uniformly at random. Player i considers switching from action a_i to the alternative $\bar{a}_i = 1 - a_i$. The transition probability for the switch is

$$T^{\text{intro}}_{\mathbf{a},\, \mathbf{a}_{i \to \bar{a}_i}} = \frac{1}{N} \big[(1 - \mu_{i0} - \mu_{i1}) \phi_i(\Delta f_i(\mathbf{a})) + \mu_{i, \bar{a}_i} \big], \tag{3}$$

where $\mu_{i,\bar{a}_i}$ denotes μ_{i0} when $\bar{a}_i = C$ and μ_{i1} when $\bar{a}_i = D$. Since ϕ_i is strictly decreasing, a larger payoff gain from keeping the current action yields a smaller switching probability. Setting $\mu_{i0} = \mu_{i1} = 0$ recovers the introspection dynamics of [1, 2]; the mutation terms add a payoff-independent probability of switching to each action.

Definition 1 (Cooperation probability)**.** *Given an ergodic chain on S with stationary distribution π, the* cooperation probability *is $p_C = \pi \cdot s$, where $s_{\mathbf{a}} = \frac{1}{N} \sum_i a_i$ is the fraction of cooperators in state $\mathbf{a}$.*

3 Individual cooperation probabilities

For additive games without mutation the stationary distribution of introspection dynamics is a product measure ([2]; Proposition 2). We show that this structure persists with mutation, with a short direct argument: for an additive game, $\Delta f_i(\mathbf{a})$ depends only on a_i, so each time player i is selected their next action is C with a fixed probability p_i regardless of the current state. The marginal cooperation probability and the product form then follow immediately.

Theorem 1 (Individual cooperation probabilities)**.** *For any additive game with constants δ_i, under introspection dynamics with player-specific selection intensity β_i and mutation probabilities $\mu_{i0}, \mu_{i1} > 0$ with $\mu_{i0} + \mu_{i1} < 1$, the long-run probability that player i cooperates is*

$$p_i = \phi_i(\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i0}. \tag{4}$$

Proof. By additivity, $\Delta f_i(\mathbf{a}) = (1 - 2a_i)\delta_i$ depends only on a_i. When player i is selected, the probability their next action is C equals p_i regardless of their current action: if $a_i = D$, the switch probability is $p_i = \phi_i(\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i0}$; if $a_i = C$, the probability of remaining at C is $1-[\phi_i(-\delta_i)(1-\mu_{i0}-\mu_{i1})+\mu_{i1}]$; using $\phi_i(x)+\phi_i(-x) = 1$, this gives $1-[(1-\phi_i(\delta_i))(1-\mu_{i0}-\mu_{i1})+\mu_{i1}] = p_i$. Since each selection of player i results in action C with probability p_i independently of their current action, the long-run cooperation probability of player i is p_i.

□

Additivity makes a player's choice forgetful: each time player i is selected they cooperate with the same probability p_i, whatever the current state. This probability is the payoff-driven Fermi value $\phi_i(\delta_i)$, scaled by the chance of no mutation and shifted by the chance of mutating to cooperation.

Theorem 2 (Stationary distribution)**.** *For any additive game with constants δ_i, the stationary distribution of the introspection chain on $S = \{0,1\}^N$ is the product measure*

$$\pi_{\mathbf{a}} = \prod_{i=1}^{N} p_i^{a_i}(1 - p_i)^{1-a_i}, \qquad \mathbf{a} \in S, \tag{5}$$

where p_i is given by (4)*.*

Proof. By Theorem 1, each selection of player i results in action C with probability p_i independently of every other player's action. In stationarity the players' actions are therefore independent, with player i cooperating with probability p_i and defecting with probability $1 - p_i$. The probability of state $\mathbf{a}$ is the product of these marginals, giving (5).

□

In the long run the players' actions are therefore independent: the group behaves as N separate biased coins, with player i showing cooperation with probability p_i. Taking the expectation of the cooperator fraction s under this product measure, the cooperation probability is the mean of the individual probabilities,

$$p_C = \frac{1}{N}\sum_{i=1}^{N} p_i. \tag{6}$$

For $\mu_{i0} = \mu_{i1} = 0$, Theorem 2 is Proposition 2 of [2]; the argument above shows that the product-measure structure is unaffected by mutation, since mutation alters only the per-player marginal p_i and not the independence between players.

Remark 1 (The two-player donation game)**.** *For the Prisoner's Dilemma M_1 of* (2)*, from [1], the payoff $f_1(\mathbf{a}) = -c_1 a_1 + b a_2$ is additive with $\delta_1 = c_1$, and similarly $\delta_2 = c_2$. Theorems 1 and 2 give the per-player cooperation probability and the product-measure stationary distribution*

$$p_i = \frac{1 - \mu_{i0} - \mu_{i1}}{1 + e^{\beta_i c_i}} + \mu_{i0}, \qquad \pi_{\mathbf{a}} = \prod_{i=1}^{2} p_i^{a_i} (1 - p_i)^{1 - a_i}.$$

Setting $\mu_{i0} = \mu_{i1} = 0$ and a common selection intensity $\beta_i = \beta$ recovers $p_i = (1 + e^{\beta c_i})^{-1}$ and

$$\pi = \frac{1}{(1 + e^{\beta c_1})(1 + e^{\beta c_2})} \big(1,\ e^{\beta c_2},\ e^{\beta c_1},\ e^{\beta(c_1 + c_2)}\big) \quad \textit{over } (CC, CD, DC, DD),$$

the stationary distribution obtained by [1] for this game; they observed that their two-player stationary distribution factorises precisely under the additive condition, of which the donation game is an instance. Mutation preserves the product form while shifting each marginal: with $b = 1$, $c_1 = 0.6$, $c_2 = 0.1$, $\beta = 5$, and asymmetric mutation $\mu_{i0} = 0.05$, $\mu_{i1} = 0.15$ we obtain $p_1 \approx 0.088$ and $p_2 \approx 0.352$, giving $\pi_{DD} \approx 0.591$, $\pi_{DC} \approx 0.321$, $\pi_{CD} \approx 0.057$, and $\pi_{CC} \approx 0.031$.

4 Structural consequences

The formula (4) for the individual cooperation probability allows immediate structural conclusions that hold for any additive game, before any specific payoff structure is imposed. We record four: the two limits of the selection intensity, a cooperation threshold, and the effect of mutation on aggregate cooperation. Without mutation, several reduce to properties already observed by [2].

Remark 2 (Neutral drift)**.** *If $\beta_i = 0$ for all i then $\phi_i \equiv \frac{1}{2}$, so the payoff parameters δ_i drop out of* (4) *and cooperation is set entirely by mutation:*

$$p_i = \frac{1 + \mu_{i0} - \mu_{i1}}{2}, \qquad p_C = \frac{1}{2} + \frac{1}{2N} \sum_{i=1}^{N} (\mu_{i0} - \mu_{i1}),$$

so $p_C = \frac{1}{2}$ exactly when $\sum_i (\mu_{i0} - \mu_{i1}) = 0$.

A player insensitive to payoff differences ignores the underlying game: their long-run behaviour is set by their mutation bias alone, and group cooperation reduces to the average asymmetry between the mutation rates.

Proposition 1 (Strong selection)**.** *For each player i,* (4) *confines the cooperation probability to the open interval $(\mu_{i0},\, 1 - \mu_{i1})$, and the endpoints are the strong-selection limits: as $\beta_i \to \infty$,*

$$p_i \to \mu_{i0} \textit{ when } \delta_i > 0, \qquad p_i \to 1 - \mu_{i1} \textit{ when } \delta_i < 0.$$

Proof. Since $0 < \phi_i(\delta_i) < 1$ and $1 - \mu_{i0} - \mu_{i1} > 0$, (4) gives $\mu_{i0} < p_i < 1 - \mu_{i1}$. As $\beta_i \to \infty$, $\phi_i(\delta_i) = (1 + e^{\beta_i \delta_i})^{-1} \to 0$ when $\delta_i > 0$ and $\to 1$ when $\delta_i < 0$, so $p_i \to \mu_{i0}$ or $p_i \to 1 - \mu_{i1}$ respectively. □

Without mutation, a unique stationary distribution requires finite β_i [2]: in the limit $\beta_i \to \infty$ the payoff-driven switches become deterministic best responses, the chain is no longer ergodic, and the long-run outcome can depend on the initial state. The strong-selection limit must therefore be taken after computing the stationary distribution at finite selection intensity, as in [1]. For example, in the donation game M_1 of (2) each $\delta_i = c_i > 0$, so as $\beta \to \infty$ the mutation-free stationary distribution collapses to a point mass on the all-defect profile. Mutation removes this degeneracy: the switching probabilities in (3) remain bounded away from 0 and 1, so the chain stays ergodic at any selection intensity and the stationary distribution converges to the non-degenerate product measure with marginals μ_{i0} or $1 - \mu_{i1}$. Mutation therefore extends the exact analysis to arbitrarily strong selection.

Corollary 1 (Cooperation threshold)**.** *Player i cooperates with probability greater than $\frac{1}{2}$ if and only if*

$$\phi_i(\delta_i) > \frac{\frac{1}{2} - \mu_{i0}}{1 - \mu_{i0} - \mu_{i1}}.$$

When $\mu_{i0} = \mu_{i1}$ this reduces to $\delta_i < 0$. A sufficient condition for $p_C > \frac{1}{2}$ is that the inequality holds for every i.

Proof. Since $1 - \mu_{i0} - \mu_{i1} > 0$,

$$p_i > \tfrac{1}{2} \iff (1 - \mu_{i0} - \mu_{i1})\phi_i(\delta_i) > \tfrac{1}{2} - \mu_{i0} \iff \phi_i(\delta_i) > \frac{\frac{1}{2} - \mu_{i0}}{1 - \mu_{i0} - \mu_{i1}}.$$

When $\mu_{i0} = \mu_{i1}$, the right-hand side equals $\frac{1}{2}$, and since ϕ_i is strictly decreasing with $\phi_i(0) = \frac{1}{2}$, this is equivalent to $\delta_i < 0$. The claim about p_C follows by averaging (6).

□

With symmetric mutation a player cooperates the majority of the time precisely when the payoff favours cooperation, $\delta_i < 0$; mutation does not move the threshold, it only softens the transition across it. With asymmetric mutation the threshold can be met even when the payoff favours defection. Substituting $\phi_i(\delta_i) = (1 + e^{\beta_i \delta_i})^{-1}$, the condition $p_i > \frac{1}{2}$ is equivalent to

$$\beta_i \delta_i < \ln \frac{\frac{1}{2} - \mu_{i1}}{\frac{1}{2} - \mu_{i0}}.$$

The right-hand side is positive precisely when the mutation is biased towards cooperation ($\mu_{i0} > \mu_{i1}$). Thus when the payoff favours defection ($\delta_i > 0$) a sufficiently strong cooperative bias still yields $p_i > \frac{1}{2}$; conversely, when the payoff favours cooperation ($\delta_i < 0$) a defection bias ($\mu_{i1} > \mu_{i0}$) can drive $p_i < \frac{1}{2}$.

Proposition 2 (Mutation-selection balance)**.** *Under common symmetric mutation $\mu_{i0} = \mu_{i1} = \mu$ for all i, with $0 \le \mu \le \frac{1}{2}$, the aggregate cooperation probability is affine in μ,*

$$p_C = (1 - 2\mu)\,\Phi + \mu, \qquad \Phi := \frac{1}{N}\sum_{i=1}^{N} \phi_i(\delta_i),$$

so that $p_C = \Phi$ at $\mu = 0$, $p_C = \frac{1}{2}$ at $\mu = \frac{1}{2}$, and

$$\frac{\partial p_C}{\partial \mu} = 1 - 2\Phi.$$

Mutation raises aggregate cooperation when $\Phi < \frac{1}{2}$ and lowers it when $\Phi > \frac{1}{2}$.

Proof. With $\mu_{i0} = \mu_{i1} = \mu$, (4) gives $p_i = (1 - 2\mu)\phi_i(\delta_i) + \mu$. Averaging over i using (6) gives the stated expression for p_C; affinity in μ, the two endpoints, and the derivative are immediate. □

Here Φ is the cooperation level of the mutation-free dynamics of [2]. Mutation interpolates linearly between this selection-driven value and neutrality, reaching $p_C = \frac{1}{2}$ as $\mu \to \frac{1}{2}$: noise erases the influence of the payoffs.

5 The heterogeneous public goods game

The *heterogeneous public goods game* [6] is played by N players with player-specific contributions $\alpha_1, \ldots, \alpha_N > 0$ and public goods multipliers $r_1, \ldots, r_N > 1$. The payoff to player i in state $\mathbf{a} = (a_1, \ldots, a_N)$ is

$$f_i(\mathbf{a}) = \frac{1}{N} \sum_{j=1}^{N} r_j \alpha_j a_j - \alpha_i a_i. \tag{7}$$

The first term is a public good shared equally by all players, in which each contribution $\alpha_j a_j$ is scaled by the contributor's multiplier r_j; the second is player i's own contribution cost. We write $P(\mathbf{a}) = \frac{1}{N} \sum_j r_j \alpha_j a_j$ for the shared pool.

The linear public goods game is additive [2], and the heterogeneous version (7), with player-specific multipliers r_i, is additive for the same reason.

Lemma 1 (Additivity of the PGG)**.** *The payoff* (7) *is additive for every player i, with*

$$\delta_i = \alpha_i \left(1 - \frac{r_i}{N}\right), \tag{8}$$

i.e. $\Delta f_i(\mathbf{a}) = (1 - 2a_i)\, \alpha_i (1 - r_i/N)$*, independent of* $\mathbf{a}_{-i}$*.*

Proof. When player i switches from a_i to $\bar{a}_i = 1 - a_i$, only their own term in the pool changes, so $P(\mathbf{a}_{i \to \bar{a}_i}) = P(\mathbf{a}) + \frac{1}{N} r_i \alpha_i (1 - 2a_i)$. Expanding both payoffs:

$$\begin{aligned} f_i(\mathbf{a}) &= P(\mathbf{a}) - \alpha_i a_i, \\ f_i(\mathbf{a}_{i \to \bar{a}_i}) &= P(\mathbf{a}) + \tfrac{1}{N} r_i \alpha_i (1 - 2a_i) - \alpha_i (1 - a_i). \end{aligned}$$

Subtracting:

$$\begin{aligned} \Delta f_i &= -\frac{r_i}{N} \alpha_i (1 - 2a_i) + \alpha_i (1 - 2a_i) \\ &= \alpha_i (1 - 2a_i) \left(1 - \frac{r_i}{N}\right). \end{aligned}$$

The pool $P(\mathbf{a})$ cancels entirely, confirming additivity with $\delta_i = \alpha_i (1 - r_i/N)$. □

The incentive to switch therefore depends only on the player's own contribution α_i and on whether their multiplier exceeds the group size: $\delta_i < 0$, an incentive to cooperate, precisely when $r_i > N$. The other players' choices never enter. We call player i a *net beneficiary* when $r_i > N$, since each unit they contribute returns more than a unit to them from the pool, and a *net contributor* when $r_i < N$, since each unit they contribute returns less than a unit.

Corollary 2 (Exact cooperation probability)**.** *For the heterogeneous public goods game* (7)*, the long-run cooperation probability is*

$$\boxed{p_C = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1-\mu_{i0}-\mu_{i1}}{1+e^{\beta_i\alpha_i(1-r_i/N)}}+\mu_{i0}\right]}. \tag{9}$$

Proof. By Lemma 1, $\delta_i = \alpha_i(1 - r_i/N)$. Substituting $\phi_i(\delta_i) = (1 + e^{\beta_i\alpha_i(1-r_i/N)})^{-1}$ into (4) and averaging by (6) gives (9).

□

Aggregate cooperation is thus the average across players of an individual logistic response to their own incentive δ_i, adjusted for mutation; no interaction terms between players survive. For $\mu_{i0} = \mu_{i1} = 0$ and a single global selection intensity, the product measure of Theorem 2 reduces to the stationary distribution of the linear public goods game given, for every state, in Proposition 3 of [2], and (9) reduces to the corresponding cooperation level.

Figure 1 validates Corollary 2 against exact values and simulation across group sizes. Table 1 verifies Theorem 2: for $N = 3$ with per-player multipliers $r_1 = 1$, $r_2 = 3 = N$, $r_3 = 9$, contributions $\alpha_1 = 1$, $\alpha_2 = 2$, $\alpha_3 = 3$, $\beta = 2$, and $\mu_{i0} = \mu_{i1} = 0.1$, the individual cooperation probabilities are $p_1 \approx 0.267$, $p_2 = 0.500$, $p_3 \approx 0.900$, and the formula and exact values of $\pi_{\mathbf{a}}$ agree to four decimal places for all eight states.

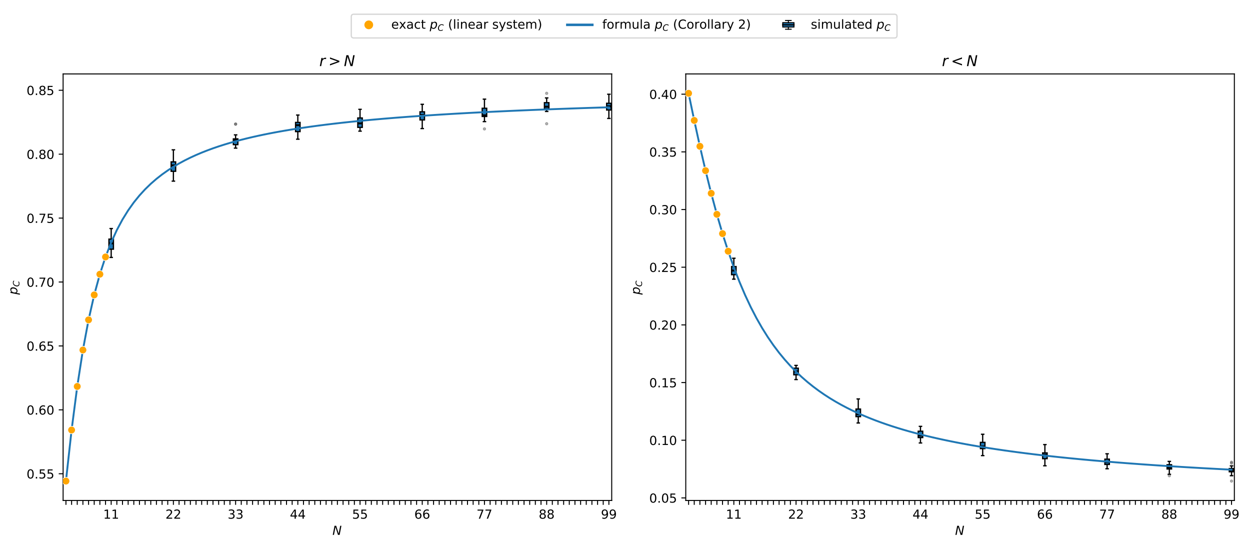

Figure 1: Validation of formula (9) on a heterogeneous group with $\alpha_i = i$, $\beta = 0.5$, $\mu_{i0} = 0.05$, $\mu_{i1} = 0.15$. *Left:* $r = 2N$ (so $r > N$ for all N). *Right:* $r = N/2$ (so $r < N$ for all N). Orange dots: exact p_C from the full 2^N linear system (feasible only for small N). Blue curve: formula (9) (Corollary 2). Blue boxes: distribution of p_C across 19 independent runs of 5,000 simulation steps each (with the first 500 steps discarded as warm-up; outliers shown). All values were obtained using [3].

We now specialise the structural consequences of Section 4 to the public goods game, where $\delta_i = \alpha_i(1 - r_i/N)$ makes every quantity explicit in the group size and the players' own parameters.

At $\beta_i = 0$ the payoff parameters drop out (Remark 2); the left panel of Figure 2 shows the curves $p_i(\beta_i)$ converging to $p_i = \frac{1}{2}$ at $\beta_i = 0$ before diverging in the direction set by r_i relative to

State $\mathbf{a}$	Formula $\pi_{\mathbf{a}}$	Exact $\pi_{\mathbf{a}}$
DDD	0.0367	0.0367
DDC	0.3299	0.3299
DCD	0.0367	0.0367
DCC	0.3299	0.3299
CDD	0.0133	0.0133
CDC	0.1201	0.1201
CCD	0.0133	0.0133
CCC	0.1201	0.1201

Table 1: Stationary-distribution probabilities $\pi_{\mathbf{a}}$ for all eight states $\mathbf{a} \in \{D, C\}^3$ with $N = 3$, $\alpha_1 = 1$, $\alpha_2 = 2$, $\alpha_3 = 3$, $\beta = 2$, $\mu_{i0} = \mu_{i1} = 0.1$, and per-player multipliers $r_1 = 1<N=r_2=3<r_3=9$. Formula: product measure (5) (Theorem 2); Exact: values from the 2^N linear system obtained using [3]. Both columns are given to four decimal places.

N.

For the PGG, $\delta_i < 0$ precisely when $r_i > N$, so the symmetric-mutation threshold of Corollary 1 becomes $r_i > N$: a player cooperates the majority of the time exactly when their personal multiplier on the public good exceeds the group size, that is, when they are a net beneficiary. Without mutation this is the cooperation-dominance condition of [2] (their $c_i < b_i/N$). The centre panel of Figure 2 illustrates this, with curves at $r_i > N$ lying above $p_i = \frac{1}{2}$ and curves at $r_i < N$ lying below it. Since the comparison is between r_i and N, a player with a fixed multiplier becomes a net contributor once the group size exceeds r_i: cooperation in large groups therefore requires multipliers that grow with N. For a net contributor ($r_i < N$, so $\delta_i > 0$) a cooperative mutation bias $\mu_{i0} > \mu_{i1}$ can still raise p_i above $\frac{1}{2}$, as in Corollary 1.

For the PGG, $\phi_i(\delta_i) > \frac{1}{2} \iff r_i > N$, so the sign of $\partial p_C/\partial\mu$ in Proposition 2 is set by whether players are, on average, net beneficiaries: common mutation raises aggregate cooperation when most players have $r_i < N$ and lowers it when most have $r_i > N$. The right panel of Figure 2 shows this mutation-selection balance for a single player: p_i is affine in the symmetric mutation rate μ (Proposition 2), interpolating between the mutation-free value $\phi_i(\delta_i)$ at $\mu = 0$ and $\frac{1}{2}$ at $\mu = \frac{1}{2}$, increasing in μ for a net contributor and decreasing for a net beneficiary.

Figure 3 isolates the role of mutation for a single player. A larger selection intensity β_i makes the player respond more sharply to payoff differences, adopting the higher-payoff action with greater probability, whereas $\beta_i = 0$ is payoff-blind random choice. Without mutation p_i is driven to 0 or 1 as β_i grows, whereas mutation confines p_i to $[\mu_{i0}, 1 - \mu_{i1}]$ (Proposition 1). The two panels show the same player ($r_i = 7$, $\alpha_i = 2$) at $N = 5$ and $N = 200$: increasing the group size alone turns the net beneficiary ($r_i > N$, p_i rising to $1 - \mu$) into a net contributor ($r_i < N$, p_i falling to μ).

Remark 3 (Role of heterogeneity)**.** *For the PGG with $r_i \neq N$, both $\partial p_i/\partial\alpha_i$ and $\partial p_i/\partial\beta_i$ are positive when $r_i > N$ and negative when $r_i < N$: each derivative carries the factor $(1 - r_i/N)$ together with a negative sign stemming from the derivative $\phi_i' < 0$, so its sign is that of $r_i/N - 1$. Indeed, β_i and α_i enter p_i only through the product $\beta_i\alpha_i$, so a change in selection intensity has the same effect as a proportional change in contribution. Under symmetric mutation, raising the mutation rate instead pulls p_i towards $\frac{1}{2}$, lowering cooperation when $r_i > N$ and raising it when $r_i < N$; with asymmetric mutation p_i is pulled towards its neutral value $(1 + \mu_{i0} - \mu_{i1})/2$.*

A net beneficiary ($r_i > N$) cooperates more as they contribute more or respond more strongly to payoff differences; a net contributor ($r_i < N$) does the opposite. Each player's long-run cooperation

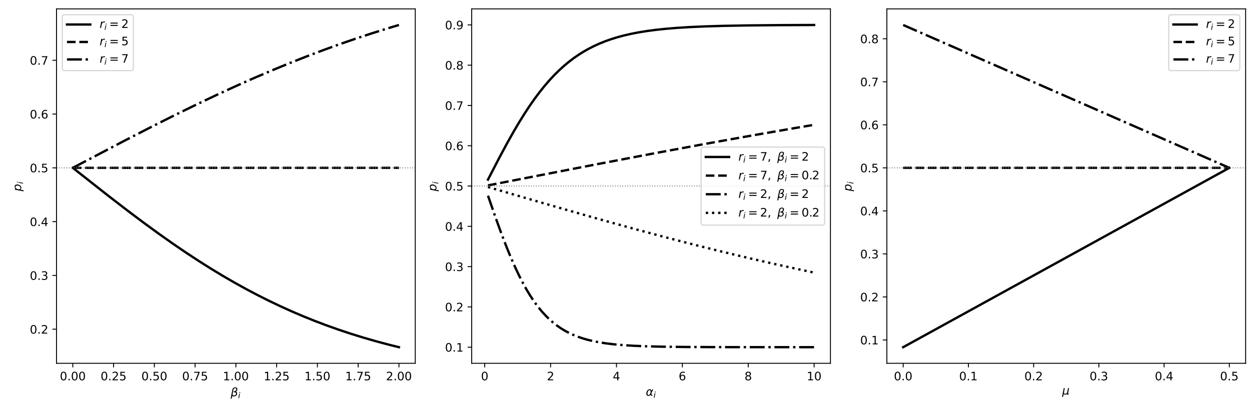

Figure 2: Structural properties of p_i for a single player with $N = 5$ and, unless varied, symmetric mutation $\mu_{i0} = \mu_{i1} = 0.1$. *Left:* p_i as a function of β_i ($\alpha_i = 2$) for $r_i < N$, $r_i = N$, and $r_i > N$; all curves pass through $p_i = \frac{1}{2}$ at $\beta_i = 0$ (Remark 2). *Centre:* p_i as a function of α_i for two values of r_i and β_i; curves above (below) $\frac{1}{2}$ correspond to $r_i > N$ ($r_i < N$). *Right:* p_i as a function of the symmetric mutation rate $\mu_{i0} = \mu_{i1} = \mu$ ($\alpha_i = 2$, $\beta_i = 2$) for $r_i < N$, $r_i = N$, and $r_i > N$; each line is affine in μ (Proposition 2) and reaches $p_i = \frac{1}{2}$ at $\mu = \frac{1}{2}$.

is thus determined entirely by their own $(\alpha_i, \beta_i, r_i, \mu_{i0}, \mu_{i1})$ and the group size N; the other players' parameters do not enter.

6 Conclusion

Building on the additive-game results of [2], we have shown that introspection dynamics with mutation on any additive game remains exactly solvable. When Δf_i depends only on a_i, each selection of player i results in action C with probability p_i regardless of the current state (Theorem 1), and the stationary distribution is a product measure (Theorem 2); the product-measure structure, due to [2] for the no-mutation case, is preserved under mutation. For the heterogeneous public goods game, the linear payoff structure implies additivity (Lemma 1), and the long-run cooperation probability has a closed form with N terms rather than 2^N (Corollary 2).

The approach does not extend to games that are not additive. When $\Delta f_i(\mathbf{a})$ depends on $\mathbf{a}_{-i}$, the per-player chains are coupled and the stationary distribution is no longer a product measure; the full 2^N linear system must be solved in general. Games such as the Stag Hunt (M_2 of (2)) contain payoff cross terms $a_i a_{-i}$ that cannot be eliminated by rescaling, so no reformulation brings them within the scope of the present results.

The source code, figures, and data for this paper are under version control and available at [4], in line with best practices for research software [8].

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We thank Marta C. Couto and Saptarshi Pal for pointing out that the product-measure decomposition for additive games was already established in [2], which an earlier version of this paper had failed to acknowledge. Harry Foster's research was supported by EPSRC grant EP/Z535126/1.

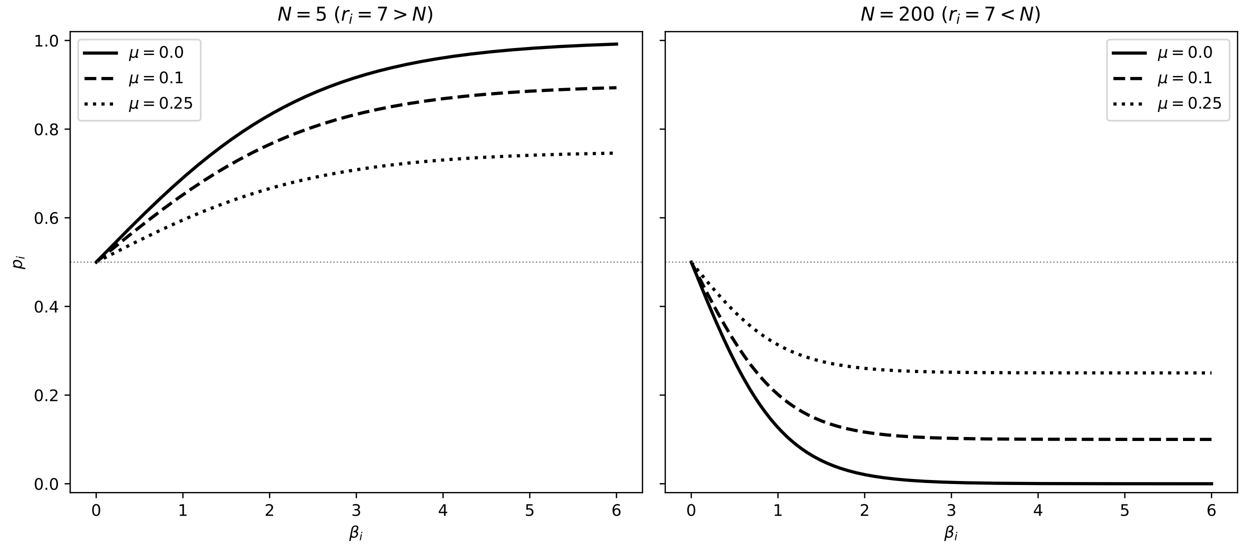

Figure 3: Effect of mutation on the cooperation probability of a single player with multiplier $r_i = 7$ and contribution $\alpha_i = 2$, as a function of the selection intensity β_i for symmetric mutation $\mu \in \{0, 0.1, 0.25\}$, at two group sizes. *Left:* $N = 5$, so $r_i > N$ and the player is a net beneficiary; p_i rises towards $1 - \mu$ as β_i grows. *Right:* $N = 200$, so the same player now has $r_i < N$ and is a net contributor; p_i falls towards μ. In both panels the $\mu = 0$ curve is the mutation-free dynamics of [2], reaching 1 or 0; mutation confines p_i to $[\mu, 1 - \mu]$. Increasing the group size alone turns the net beneficiary into a net contributor.

References

[1] Marta C. Couto, Stefano Giaimo, and Christian Hilbe. Introspection dynamics: a simple model of counterfactual learning in asymmetric games. *New Journal of Physics*, 24(6):063010, 2022.

[2] Marta C. Couto and Saptarshi Pal. Introspection dynamics in asymmetric multiplayer games. *Dynamic Games and Applications*, 13:1215–1246, 2023.

[3] Harry Foster and Vince Knight. hefos/ludics: v0.1.1, April 2026.

[4] Harry Foster and Vince Knight. introspection-dynamics-with-mutation-in-additive-games. `https://github.com/hefos/introspection-dynamics-with-mutation-in-additive-games`, 2026.

[5] Drew Fudenberg, Martin A. Nowak, Christine Taylor, and Lorens A. Imhof. Evolutionary game dynamics in finite populations with strong selection and weak mutation. *Theoretical Population Biology*, 70:352–363, 2006.

[6] Oliver P. Hauser, Christian Hilbe, Krishnendu Chatterjee, and Martin A. Nowak. Social dilemmas among unequals. *Nature*, 572:524–527, 2019.

[7] Martin A. Nowak. Five rules for the evolution of cooperation. *Science*, 314:1560–1563, 2006.

[8] Greg Wilson, Jennifer Bryan, Karen Cranston, Justin Kitzes, Lex Nederbragt, and Tracy K. Teal. Good enough practices in scientific computing. *PLOS Computational Biology*, 13(6):e1005510, 2017.